

COMM114: Introduction

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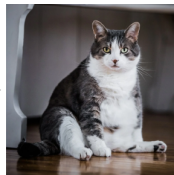
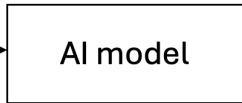
September 2025

Definition of Generative AI

Generative artificial intelligence (Generative AI) is a subfield of artificial intelligence that uses **generative models** to produce text, images, videos, or other forms of data. These models learn the underlying **patterns and structures** of their **training data** and use them to produce **new data** based on the input, which often comes in the form of natural language prompts. (from wikipedia).

Prompt:

“a picture of cat”



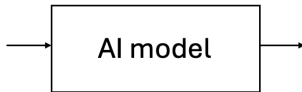
New sample

Definition of Generative AI (cont.)

Generative artificial intelligence (Generative AI) is a subfield of artificial intelligence that uses **generative models** to produce text, images, videos, or other forms of data. These models learn the underlying **patterns and structures** of their **training data** and use them to produce **new data** based on the input, which often comes in the form of natural language prompts. (from wikipedia).

Prompt:

"A ballet dancer twirls
on the surface of a still
lake at sunset"

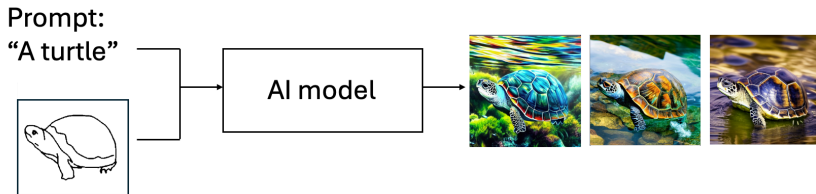


[a video sample](#)

Remark: The link for the video sample is [here](#).

Definition of Generative AI (cont.)

- The input to the AI model could also be in other formats such as text-prompt + sketch.
- For the same input, the model could generate different flavors of images (we will explain it later).



From Generation to Sampling

We represent images and videos as vectors

- Images

- Height H and Width W
- 3 color channels (RGB)
- $\mathbf{x} \in \mathbb{R}^{H \times W \times 3}$

- Videos

- T times frames
- Each frame is image
- $\mathbf{x} \in \mathbb{R}^{T \times H \times W \times 3}$



Vector representation

The objects we want to generate are represented as vectors:

$$\mathbf{x} \in \mathbb{R}^d$$

From Generation to Sampling (cont.)

What is meaning of successfully generating new data?

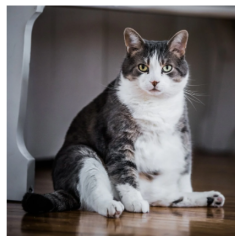
Prompt: "A picture of cat"



Bad



Wrong animal



Good

Question:

Can we formalize the process of generating **good** new sample given the input prompt?

From Generation to Sampling (cont.)

- Intuitively speaking, a good new sample is successfully generated by a AI model if it looks as if it came from the data distribution $p_{data}(\mathbf{x})$.

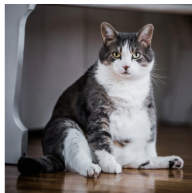
Prompt: "A picture of cat"



Bad



Wrong animal



Good

- For the example above, $p_{data}(\mathbf{x})$ models the distribution of images of all kinds of cats. The images below are samples drawn from $p_{data}(\mathbf{x})$:



$\mathbf{x}_1$



$\mathbf{x}_2$



$\mathbf{x}_3$

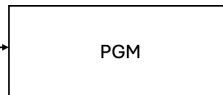
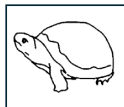
...

From Generation to Sampling (cont.)

- An AI model is in fact a probabilistic generative model (PGM).
- **Definition of PGM:** A probabilistic generative model (PGM) is a probability distribution over a random variable $\mathbf{X}$ which attempt to learn from a set of observed data $\{\mathbf{x}_i \sim p_{data}(\mathbf{x})\}_{i=1}^n$ with some probability density $p_{\theta}(\mathbf{x})$, which is parameterized by θ .
- Generation of new samples from an AI model is in fact sampling from a PGM $p_{\theta}(\mathbf{x})$.

Prompt:

“A turtle”



different samples

From Generation to Sampling (cont.)

- In the ideal case, if the PGM $p_{\theta}(\mathbf{x})$ is a good approximation of $p_{data}(\mathbf{x})$ (i.e., $p_{\theta}(\mathbf{x}) \approx p_{data}(\mathbf{x})$), the generated samples by the PGM look as if they came from the data distribution.

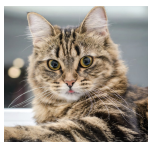
Dataset for Training AI models

- We assume that the observed data samples $\{\mathbf{x}_i\}_{i=1}^n$ are drawn from the data distribution $p_{data}(\mathbf{x})$:

$$\mathbf{x}_i \sim p_{data}(\mathbf{x}), \quad i = 1, 2, \dots, n$$



$\mathbf{x}_1$



$\mathbf{x}_2$



$\mathbf{x}_3$

...

- $p_{data}(\mathbf{x})$ is a probability density function if
 - p_{data} is a function that maps from $\mathbb{R}^d$ to $\mathbb{R}$.
 - $p_{data}(\mathbf{x}) \geq 0$ for all $\mathbf{x} \in \mathbb{R}^d$
 - $\int_{\mathbf{x}} p_{data}(\mathbf{x}) d\mathbf{x} = 1$
- The objective of any PGM is to approximate $p_{data}(\mathbf{x})$ by training the model with the given data samples $\{\mathbf{x}_i\}_{i=1}^n$.
- **Note:** In practice, most of the time, we don't know $p_{data}(\cdot)$

Dataset for Training AI models (cont.)

Example of dataset sources:

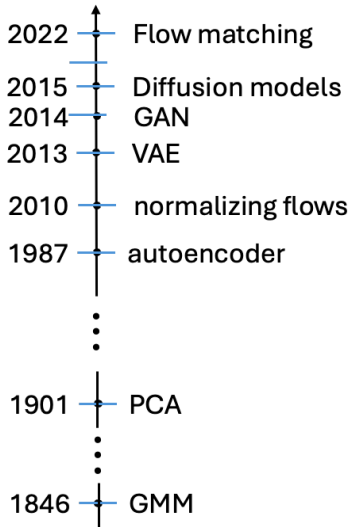
- Images: Publicly images from Internet
- Video: Youtube, Youku, and so on.
- Audio: Youtube, youku, and so on.

Without loss of generality, we assume that a dataset consists of a finite number of observed samples from the data distribution:

$$\mathbf{x}_i \sim p_{data}, \quad i = 1, \dots, n.$$

Generative AI history

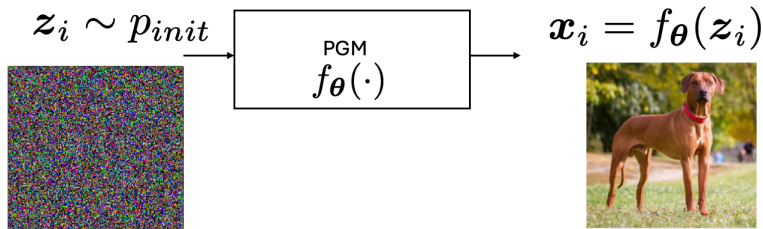
Different distributions and different learning objectives and approaches lead to different PGMs (or generative AI models), e.g., GANs, VAEs, NFs etc.



Generative AI history (cont.)

Classification of PGMs

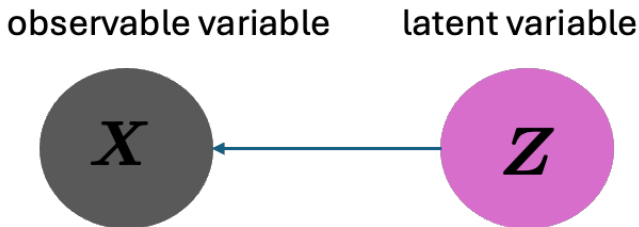
- Input data: $\{\mathbf{x}_i\}_{i=1}^n$
- Some PGMs **directly** learns $p_{\theta}(\mathbf{x})$ from the dataset $\{\mathbf{x}_i\}_{i=1}^n$.
 - Gaussian mixture model (GMM), normalizing flows (NFs)
- Other PGMs **implicitly** learns $p_{\theta}(\mathbf{x})$ from the dataset $\{\mathbf{x}_i\}_{i=1}^n$. In particular, they learn to convert samples from an initial distribution (e.g., Gaussian distribution) into samples from the data distribution.



- In the diagram above, the random variable $\mathbf{z}$ is referred to as the latent variable.

Why introducing latent variable in the 2nd class of PGMs?

- p_{init} is a simple parametric distribution (e.g., Gaussian distribution), from which one can easily draw samples. With p_{init} , the PGMs only need to focus on learning the mapping function f_{θ} .
- The mapping function f_{θ} implicitly transforms $p_{init}(z)$ to $p_{\theta}(x)$ where $p_{\theta}(x) \approx p_{data}(x)$.
- Mathematically, we introduce observable random variable X for the data samples and the latent variable Z for the initial distribution.



Why introducing latent variable in the 2nd class of PGMs?

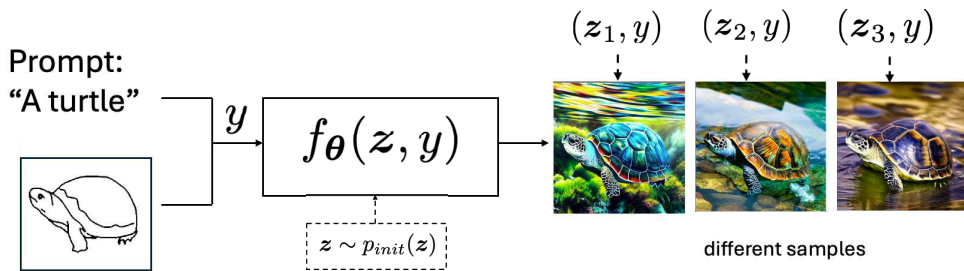


Figure: Demonstration of image generation for the 2nd class of PGMs

Remark: For the same input y (e.g., including both the text prompt and the sketch), different noise samples $\{z_1, z_2, z_3\}$ drawn from $p_{init}(\cdot)$ generate different flavors of images.

Demonstration of fake face generation

Try the demo on website of [fake face generation](#)

Module Evaluation

- Coursework 1 (30%)
 - python notebook tasks
- Coursework 2 (70%)
 - part 1: python notebook tasks (20% of total mark)
 - part2: a small project (50% of total mark)

References

- Lecture notes from MIT on Flow Matching and Diffusion Models

Questions?